TEXORA · LECTURE SHEET

Folding and Packing

Board folds, stand-up folds and hanger packs; tagging and polybagging; solid and ratio carton packing with marking and packing lists; and the packing audit that releases a shipment.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 9 · LEVEL 3 TERM 1

1. Packing is part of the product

The customer meets the garment folded, tagged and bagged; a creased, mislabelled or damp-packed garment is judged defective before it is unfolded. Packing also has an engineering content: fold geometry protects the pressed finish, carton packing protects the shipment, and the packing documents (ratio, assortment, carton marking) are how the buyer's warehouse verifies 50,000 pieces without opening every box. Every step follows the buyer's **packing manual** for the style; where the manual is silent, the factory standard below applies.

2. Folding methods

METHOD	USED FOR	NOTES
Board (template) fold	Shirts, T-shirts, polos	Garment folded around a board of the buyer's set dimensions (e.g. 24 × 34 cm for a men's shirt) giving a uniform rectangle; board withdrawn, fold held by clips/pins or the bag.
Stand-up (collar) fold	Dress shirts	Collar supported upright with a butterfly/plastic support, band and tissue at the neck; premium presentation.
Hanger pack (garment on hanger, GOH)	Suits, dresses, coats; also container GOH	No folding at all: garment hung, covered with a polybag, shipped hanging or in hanging cartons.
Roll / vacuum pack	Knitwear, travel and promotional goods	Air removed to cut volume; only for crease-tolerant goods — the finish trade-off is explicit.
Flat pack with tissue and support board	Delicate and premium flat goods	Tissue interleaving prevents crease-setting and dye transfer between layers.

Folding discipline: fold on a clean table, buttons fastened per manual (usually all, or alternate), size sticker and fold position consistent so a stack reads uniformly; never fold a warm or damp garment — trapped moisture mildews in transit (the container voyage to Europe can take five weeks), so garments are cooled after pressing and packing rooms control humidity; silica gel is added where the manual requires it.

3. Tagging, bagging and carton packing

- **Tags** — hang tag (brand/price) attached with a tag gun or string at the manual's position (never through a visible face fabric where the needle mark will show); spare-button bag and care information included as specified.
- **Polybag** — size printed or stickered, suffocation-warning text as destination law requires, vent holes where specified; bag thickness and recycled content increasingly specified by buyers' sustainability rules. One garment per bag (single pack) or several (multi-pack) per manual.
- **Assortment (ratio pack)** — cartons are packed either **solid** (one size/colour per carton) or in **ratio assortment** (e.g. a 12-piece carton holding S:M:L:XL = 2:4:4:2, matching the order ratio), and pre-packs for retail are built exactly to the store's ratio.
- **Cartons** — standard export cartons (commonly about 60 × 40 × 40 cm class sizes), gross weight usually capped near 15–20 kg by buyer rule and handler safety; carton strength (ply and burst rating) matched to stacking in the container.
- **Carton marking** — printed on two sides: buyer, style, PO number, colour, size breakdown, quantity, carton number of total, gross/net weights, dimensions, destination and handling marks. The packing list per carton and the summary packing list reconcile carton-by-carton with the order.

4. A worked packing calculation

Example. Order: 24,000 shirts in ratio S:M:L:XL = 1:2:2:1, packed in ratio cartons of 24 pieces. Ratio parts = 6; one part = 4 pieces per carton, so each carton holds S 4, M 8, L 8, XL 4. Cartons = 24,000 ÷ 24 = **1,000 cartons**. At 0.45 kg per packed shirt, net = 10.8 kg; with carton and bags ≈ 12 kg gross — inside a 15 kg cap, so the 24-piece carton stands. If the cap were 10 kg, the carton would drop to 18 pieces and the count would rise to 1,334 cartons: packing arithmetic, like thread arithmetic (Topic 3), is simple — and a wrong figure is a warehouse dispute.

5. Final audit before dispatch

Before cartons are sealed for shipment, the factory runs a **packing audit**: random cartons are opened and checked for garment count, size ratio, folding standard, bag and tag correctness and carton marking, and the shipment is metal-detected where the buyer requires it (needle-control policy: broken-

needle log, detection of all garments for children's wear). Only audited, reconciled cartons are released to the C&F agent.

Glossary

TERM	MEANING
Ratio assortment	Packing sizes in one carton or pre-pack in the proportions of the order ratio.
GOH	Garment on hanger: shipment without folding, hanging in cartons or in a fitted container.
Packing list	The per-carton and summary documents reconciling contents with the purchase order.
Needle control	The factory system of logging broken needles and metal-detecting garments to guarantee no fragment ships.

Key points and summary

- Fold geometry, tagging, bagging and carton data are all specified in the buyer's packing manual — packing is manufacture.
- Never pack warm or damp: trapped moisture becomes mildew during the voyage.
- Cartons are packed solid or by ratio; marking and packing lists reconcile the shipment piece by piece.
- A packing audit and (where required) metal detection close the process before dispatch.

Practice

Answer in your own words; brief model answers follow.

1. An order of 18,000 pieces is packed 30 per carton in ratio 1:2:2:1. How many cartons, and how many of each size per carton?
2. Why must pressing and packing be separated by a cooling interval, and what defect appears if they are not?
3. A buyer caps carton gross weight at 12 kg; a packed garment weighs 0.38 kg and the empty carton with bags is 1.1 kg. What is the largest whole number of garments per carton?

Answer key

1. Cartons = 18,000 ÷ 30 = 600. Ratio parts = 6, one part = 5 pieces: S 5, M 10, L 10, XL 5 per carton.

2. A warm/damp garment sealed in a polybag traps moisture; over a multi-week voyage this produces mildew spots and odour — an unrecoverable defect. Cooling (and humidity control) lets the garment reach equilibrium before bagging.
3. Available for garments = 12 − 1.1 = 10.9 kg ÷ 0.38 = 28.7, so 28 garments per carton.

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